

Example 2: Fetch data from Statistics Estonia with R

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1 Setup R

```
library(quarto)      # for compiling Quarto presentations and documents
library(pxweb) # query and fetch data from Statistics Estonia
library(tidyverse)  # dplyr, ggplot2, readr, etc.
library(here)   # path settings
```

2 Accessing Statistics Estonia Data with pxweb

- [Statistics Estonia](#) is not yet part of DBnomics
 - There is **no dedicated R package** for [andmed.stat.ee](#)
 - **However**, you can access the data using the `pxweb` R package
 - It is based on the **PX-WEB API**, also used by many Nordic statistical agencies
-

2.1 Data Access Options

- **Interactive Access**
Browse tables and build queries via the Statistics Estonia website in the R console
- **Direct Programmatic Query**
Use the API endpoint (a *URL*) and supply **key-value pairs** to specify:
 - Variables (e.g., indicators, time, sector)
 - Values or selections for each variable

3 Find and download data interactively

```
pxweb_interactive() # All available databases
pxweb_interactive("https://andmed.stat.ee/api/v1/en/stat") # Statistics Estonia
```

```
#| eval: false
```

```
=====  
R PXWEB: Content of 'andmed.stat.ee'  
      at '/api/v1/en/stat'
```

```
=====  
[ 1 ] : Environment  
[ 2 ] : Economy  
[ 3 ] : Population  
[ 4 ] : Social life  
[ 5 ] : Multidomain statistics  
[ 6 ] : Population and Housing Census  
[ 7 ] : Discontinued datasets  
=====
```

```
Enter your choice:
```

```
('esc' = Quit, 'b' = Back, 'i' = Show id)
1:
```

4 GET call to PXWEB API

```
pa111_meta <- pxweb_get(
  url = "https://andmed.stat.ee/api/v1/en/stat/PA111")
pa111_meta
```

PXWEB METADATA

PA111: AVERAGE MONTHLY GROSS WAGES (SALARIES), MEDIAN, DECILES AND NUMBER OF EMPLOYEES BY ECONOMIC

variables:

```
[[1]] Näitaja: Indicator
[[2]] Tegevusala: Economic activity
[[3]] Vaatlusperiood: Reference period
```

The object is just metadata, no data yet!

How to identify key-value pairs?

- Use either:
 1. Programmatically via the metadata (`pxweb_get()`)
 2. From the Statistics Estonia web interface – click “**API query for this table**”

5 Key-value pairs

- Values for the first key (Näitaja)

```
pa111_meta$variables[[1]]
```

\$code

```
[1] "Näitaja"
```

\$text

```
[1] "Indicator"
```

\$values

```

[1] "GR_W_AVG"      "NR_EMPL"      "GR_W_D1"      "GR_W_D2"      "GR_W_D3"
[6] "GR_W_D4"      "GR_W_D5"      "GR_W_D6"      "GR_W_D7"      "GR_W_D8"
[11] "GR_W_D9"      "GR_W_AVG_SM"

```

```
$valueTexts
```

```

[1] "Average monthly gross wages (salaries), euros"
[2] "Number of employees"
[3] "1st decile of monthly gross wages (salaries), euros"
[4] "2nd decile of monthly gross wages (salaries), euros"
[5] "3rd decile of monthly gross wages (salaries), euros"
[6] "4th decile of monthly gross wages (salaries), euros"
[7] "Median (5th decile) of monthly gross wages (salaries), euros"
[8] "6th decile of monthly gross wages (salaries), euros"
[9] "7th decile of monthly gross wages (salaries), euros"
[10] "8th decile of monthly gross wages (salaries), euros"
[11] "9th decile of monthly gross wages (salaries), euros"
[12] "Percentage change in average gross wages (salaries) compared to same period in previous

```

```
$elimination
```

```
[1] FALSE
```

```
$time
```

```
[1] FALSE
```

6 Fetch data

- Set: Key-value pair (as R list object) & Url to the database
- Wildcard operator * to select all values

```

px_query_list1 <- list(
  "Näitaja" = c("GR_W_D1", "GR_W_D5", "GR_W_D9"),
  "Tegevusala" = "*",
  "Vaatlusperiood" = "*"
)

pa111 <- pxweb_get(
  url = "https://andmed.stat.ee/api/v1/en/stat/PA111",
  query = px_query_list1
)

```

- We get a list object with the data

```
pa111
```

PXWEB DATA

With 4 variables and 960 observations.

```
class(pa111)
```

```
[1] "pxweb_data" "list"
```

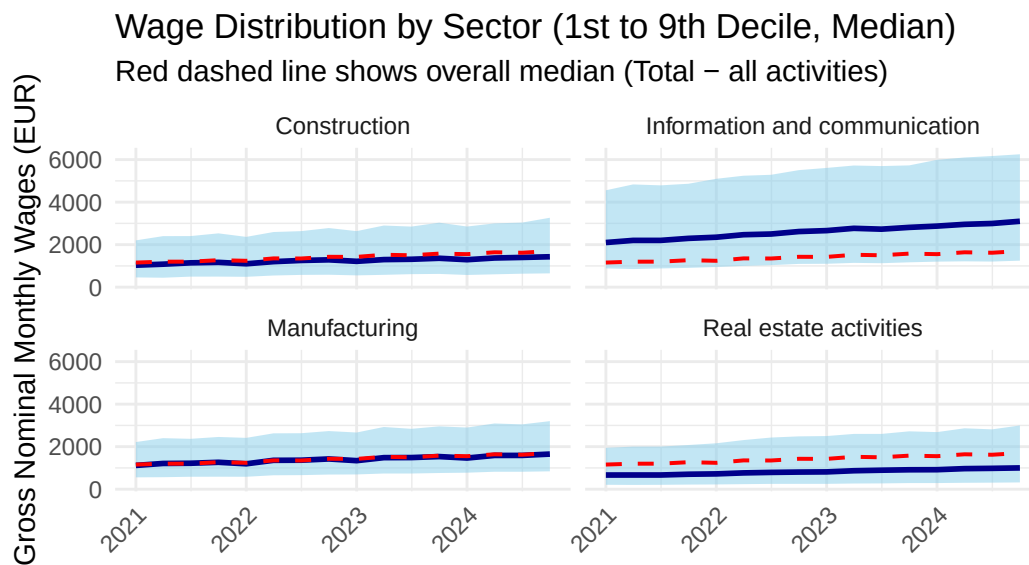
7 Convert the pxweb list object to a data frame

```
# Convert to data frame
pa111.df <- as.data.frame(pa111, column.name.type = "text", variable.value.type = "text")
pa111.code <- as.data.frame(pa111, column.name.type = "code", variable.value.type = "code")
# Optional: Combine text and code in one data frame
pa111.df$Näitaja <- pa111.code$Näitaja
```

```
head(pa111.df, n=3)
```

	Indicator	Economic activity
1	1st decile of monthly gross wages (salaries), euros	Total - all activities
2	1st decile of monthly gross wages (salaries), euros	Total - all activities
3	1st decile of monthly gross wages (salaries), euros	Total - all activities
	Reference period	
1	2021 Q1	
2	2021 Q2	
3	2021 Q3	
	PA111: AVERAGE MONTHLY GROSS WAGES (SALARIES), MEDIAN, DECILES AND NUMBER OF EMPLOYEES	
1		532
2		472
3		508
	Näitaja	
1	GR_W_D1	
2	GR_W_D1	
3	GR_W_D1	

```
ggplot(pa111_plotdata, aes(x = quarter)) +
  geom_ribbon(aes(ymin = D1, ymax = D9), fill = "skyblue", alpha = 0.5) +
  geom_line(aes(y = Median), color = "darkblue", linewidth = 1) +
  geom_line(aes(y = total_median), color = "red", linetype = "dashed", linewidth = 0.7) +
  facet_wrap(~Economic activity`, ncol = 2) +
  labs(
    title = "Wage Distribution by Sector (1st to 9th Decile, Median)",
    subtitle = "Red dashed line shows overall median (Total – all activities)",
    caption = source_note,
    x = "",
    y = "Gross Nominal Monthly Wages (EUR)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Source: Statistics Estonia
 DECILES AND NUMBER OF EMPLOYEES by Indicator, Economic activity and Reference period"

8 Rendering All Formats from R or Bash

To render the document to **all output formats** (docx, Beamer, Reveal.js): run programmatically from another script or the R console:

```
quarto::quarto_render(here("teaching_materials", "example2_statistics_estonia.qmd"), output_
```

or from the command line:

```
quarto render example2_statistics_estonia.qmd
```

9 End of session